
PhyGround

Benchmarking Physical Reasoning
in Generative World Models

Looks real. Breaks gravity.

Visual fidelity $\neq$ physical correctness

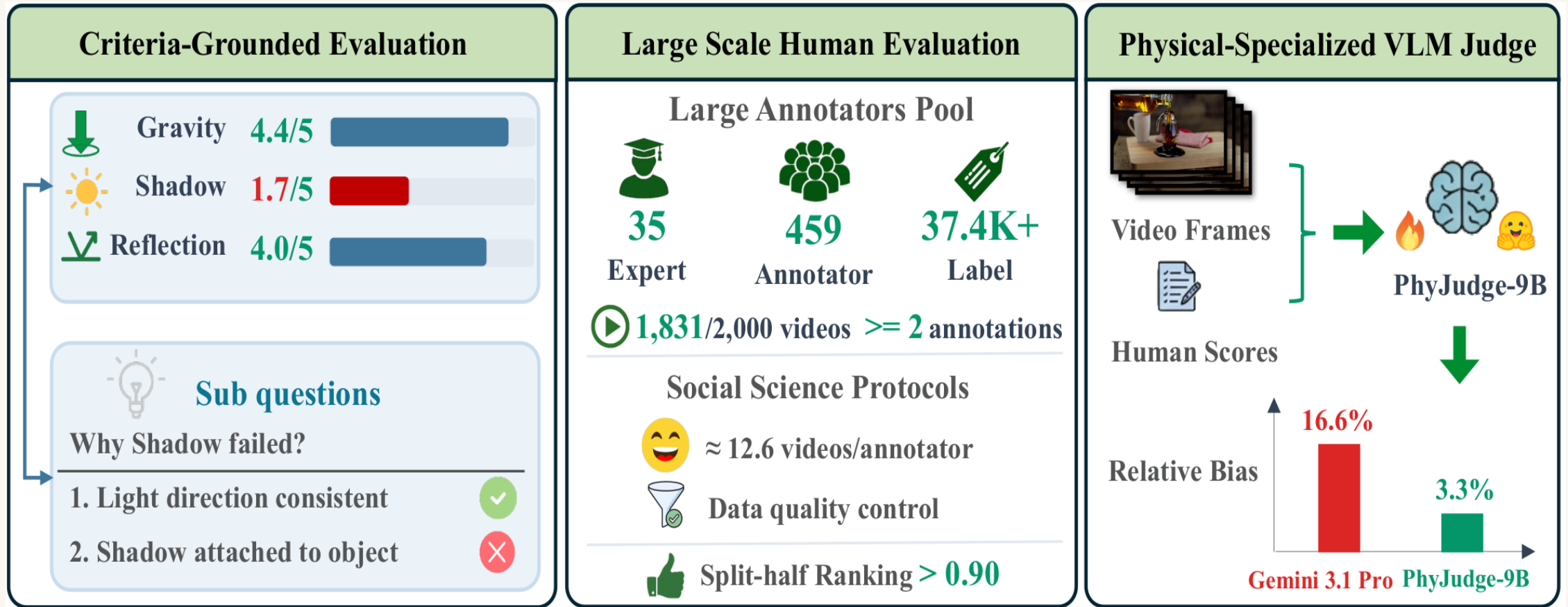
floating objects • vanishing fluids • broken shadows

Why current physics evaluation falls short

- **Coarse:** domain-level scores hide which law fails
- **Biased:** small annotator pools, fatigue, response shortcuts
- **Opaque:** closed-source VLM judges; FVD/SSIM ignore physics

We need diagnostics, not a single number.

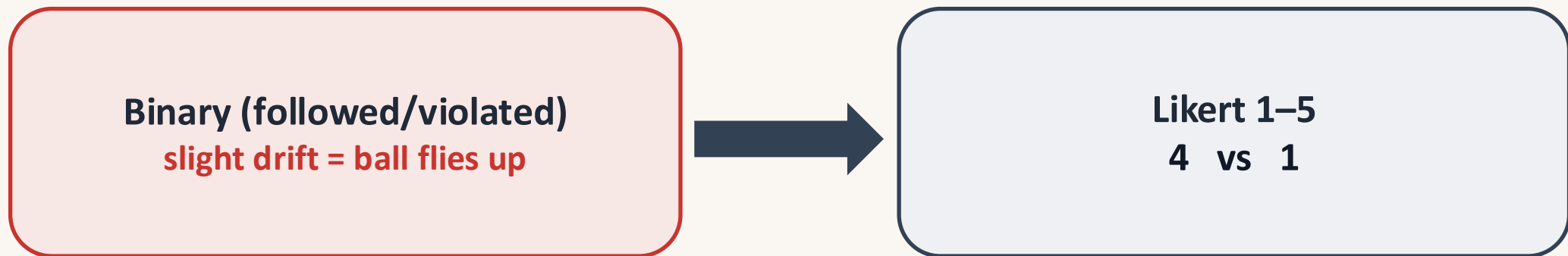
PhyGround in one picture



Per-law scoring → quality-controlled human study → open VLM judge

Per-law diagnostics, not one number

- Each video scored on **2–4 applicable laws**
- **1–5 Likert** preserves severity
- **Sub-questions** anchor what evaluators look at



Same overall score, different physics failures

Model	Overall	Solid	Fluid	Optical
Wan2.2-27B-A14B	3.28	3.23	3.18	3.55
Veo-3.1†	3.28	3.12	3.65	3.69

- Identical Overall → **opposite** domain strengths
- Wan2.2 wins solid-body • Veo-3.1 wins fluid & optical
- **One number hides the deployment-relevant choice**

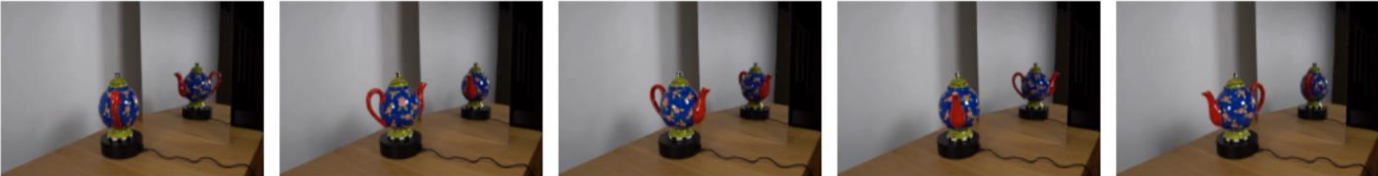
PhyJudge-9B: an open, reproducible judge

Aggregate relative bias

3.3% vs 16.6%

PhyJudge-9B | Gemini-3.1-Pro

- Open weights, no API drift
- Trained on human labels
- Held out: no Veo-3.1 in training



Domain: Optical Model: Veo3.1

Prompt: A teapot on a rotating display base in front of a mirror; the reflection rotates in the opposite direction.

Judge	SA	PTV	Persist.	Reflection	Shadow	Overall
Human	4.0	4.3	4.3	4.7	4.3	4.4
PhyJudge	4	5	5	4	5	4.6
Gemini	3	5	2	1	1	2.2

Gemini-3.1-Pro reasoning (verbatim, conclusion sentence):

Reflection (Gemini=1, Human=4.7):
“when the real spout points towards the mirror, the reflected spout points away from it, violating the basic laws of reflection. The physical behavior is severely incorrect”
Shadow (Gemini=1, Human=4.3):
“The shadow of the teapot is baked into the object/base and rotates with it, completely violating the stationary light source”

Veo-3.1 rotating-mirror teapot — head-to-head

Takeaway

- Physics evaluation needs **per-law diagnostics**, not one number
- Quality-controlled human studies are **reproducible**
- Open judges close the **auditability** gap

*If you can't say which law a model breaks,
you can't fix it.*